

SOURCES OF INFECTION AND METHODS OF CONTROL OF *SEPTORIA OENOTHERAE* IN EVENING PRIMROSE (*OENOTHERA* SPP.)

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Septoria oenotherae West. can cause severe damage in overwintered crops of evening primrose (*Oenothera* spp.), including complete crop loss. Damage would be reduced if the sources of infection could be identified and removed. Examination of seed capsules inoculated with *S. oenotherae* showed that 96% of the pycnidia present were on the outside of the capsules, and seeds bearing pycnidia were only rarely found. However, internal infection of seeds from these capsules was demonstrated by both a blotter test and by culturing on agar media. Immersing seeds in 45°C water for 25 minutes destroyed viable fungal propagules located internally in seeds without reducing seed germination. The pathogen was also shown to overwinter in the pycnidial stage on stems left standing in the field. It is concluded that both internal seed-borne infection and overwintered crop debris are potential sources of infection in commercial seed stocks of evening primrose.

Key words: evening primrose, *Oenothera* spp., *Septoria oenotherae*, crop loss, seed-borne infection, hot water treatment

Introduction

Evening primrose (*Oenothera* spp.) is a high-value oilseed crop that is grown in temperate areas of northern and central Europe, North America and Australasia. It is a monocarpic biennial and the seeds are formed in capsules that are arranged in a spike. The seed oil is an important source of γ -linolenic acid, which has valuable nutritional and pharmaceutical properties (Horrobin, 1992). The crop may be either overwintered or spring-sown. Overwintered crops bolt more reliably, have fewer harvesting problems owing to their earlier maturity (Fieldsend and Morison, 2000a), and have higher seed oil contents (Fieldsend and Morison, 2000b), but are susceptible to damage by *Septoria oenotherae* West., which may result in complete crop loss (Simpson et al., 1995).

S. oenotherae was first reported to be seed-borne in evening primrose by Rawlinson and Dover (1986). Simpson et al. (1995) found that almost 75% of commercial seed stocks contained pycnidia of *S. oenothera* and that these were normally found on capsule, stem or leaf fragments and only rarely on the seeds. These pycnidia contained viable conidia and test seedlings of evening primrose became infected following inoculation with these conidia. Stobart et al. (1999) demonstrated a positive relationship between the number of pycnidia per gram of seed sown and the initial disease level in the field.

Separation of debris from infected seed stocks is difficult because evening primrose seeds are small (1 000 seeds weigh 0.4–0.5 g) and irregular in shape (Simpson et al., 1995). Furthermore the effectiveness of this control method would be reduced if *S. oenotherae* infection could be carried within the seeds. Stobart et al. (1999) observed *S. oenotherae* even in plots sown with seed free of pycnidia and suggested internal seed-borne infection as a source. This paper reports that internal *S. oenotherae* infection can be detected in evening primrose seeds and that hot water treatment is an effective method of control.

A further strategy for reducing the levels of *S. oenotherae* is to recommend a period of rotation between evening primrose crops. Simpson et al. (1995) have shown that the pathogen could survive when infected debris was buried in the soil for ten months. This paper demonstrates the potential of crop debris left standing in the field as a source of infection to subsequent crops.

Materials and methods

Plant material

In 1995, and again in 1996, 25 plants of an evening primrose breeding line known to be highly susceptible to *S. oenotherae* were raised under glass until the rosettes were approximately 100 mm in diameter, then planted out in April in a sheltered garden site in Writtle, Essex, UK. The plants were spaced 150 mm apart to encourage disease spread. Infected leaves collected from field crops were placed on the rosettes to inoculate them with *S. oenotherae*. Immediately after flowering an infected leaf was tied to the top of each flower spike to encourage capsule and seed infection. On dry days the plants were watered at dusk to facilitate disease spread. In October of both years capsules with high numbers of *S. oenotherae* pycnidia on their outer surfaces were individually harvested and allowed to dry.

Distribution of S. oenotherae pycnidia on seeds and capsules

The numbers of pycnidia on the lower and upper halves of 121 capsules harvested in 1995 were recorded using a stereo microscope (magnification $\times 20$). Subsequently each capsule was opened and the number of pycnidia on the internal surface, the placenta and the seeds was counted. The percentage distribution of pycnidia was then calculated for each surface.

Assessment of internal infection in hand-harvested seeds

Ten heavily-infected capsules were harvested from the plants grown in 1996 and the distribution of pycnidia was again recorded. Seeds void of pycnidia were selected for assessment of internal infection based on a blotter test and an agar culture test as outlined for *S. nodorum* on *Triticum aestivum* (Anonymous, 1993). Forty seeds from each capsule were surface sterilised with 10% sodium hypochlorite for 3 min and then rinsed in sterile distilled water. For the blotter test, 90 mm diameter filter papers (Whatman 1, Merck Ltd., Dorset, UK) were set in the bottom and lid of a 100 mm Petri dish and wetted with 3 and 1 ml distilled water, respectively. Ten seeds were set in the prepared Petri dish, which was sealed with parafilm and incubated under fluorescent lights at 20°C for 21 days. The developing seedlings were checked for the presence of pycnidia at weekly intervals.

A further 40 seeds from each of the ten capsules were surface-sterilised, rinsed and transferred aseptically onto potato dextrose agar (PDA) (Merck Ltd., Dorset, UK). These Petri dishes were incubated as before and developing fungal colonies were recorded and identified over a 14-day period. Samples from colonies thought to be *S. oenotherae* were used to inoculate healthy evening primrose seedlings grown on cylindrical cellulose plugs as described by Simpson et al.

(1995). These were placed under fluorescent lights and the development of disease symptoms was recorded over 21 days. Larger pot-grown plants with rosette diameters of 200 mm and showing no disease symptoms were also inoculated. Agar from the leading edge of the colonies was placed on leaves at sites that had been pierced with a sterile needle to encourage infection. The plants were placed inside a large polyethylene bag and left on a north-facing window ledge for 21 days. After the completion of the PDA tests a dilute V8 agar medium, as used by Reeleder (1994) and an agar medium devised for the sporulation of *S. nodorum* (Richards, 1951) were used to culture *S. oenotherae* from infected evening primrose seeds.

Assessment of infection in combine-harvested seed stocks

A commercial seed crop of evening primrose cv. Rigel specially grown under a cropping regime designed to minimise *S. oenotherae* infection was harvested in October 1995. A 4 g sample of seeds was checked for the presence of pycnidia on seeds and debris. Following surface sterilising and rinsing, 400 seeds were plated onto PDA as described above. A further seed sample was harvested in autumn 1996 from untreated control plots in a fungicide trial. The incidence and severity of *S. oenotherae* in these plots was greater than would normally be expected in commercial crops (100% and 63%, respectively, six months prior to harvest). Agar and blotter tests were carried out using seeds from this sample.

Control of seed-borne S. oenotherae using hot water treatment

Four hundred pycnidia-free seeds from each of ten commercial crops of evening primrose cv. Rigel harvested in the Netherlands in 1997 and 1998 were tested for their germination (after 7 and 14 days) and *S. oenotherae* infection (after 21 days) percentages in April 2000 using the blotter test method previously described.

One of these seedlots was then used for the hot water treatment studies, using a method described by Bant and Storey (1952). For each temperature treatment, 400 seeds were enclosed in a muslin cloth and immersed in water maintained at the required temperature (ranging from 45°C to 58°C) for 25 minutes, then allowed to dry. After treatment the germination ability of the seeds and the level of infection with *S. oenotherae* were again measured using the blotter test method.

In all tests, 25 seeds were set per Petri dish, giving 16 Petri dishes per treatment.

Overwintering of S. oenotherae on aerial plant debris

The stems and remaining capsules of the inoculated test plants were left to overwinter and in April 1997 pycnidia containing conidia thought to be those of *S. oenotherae* were used to inoculate healthy evening primrose seedlings growing on cylindrical cellulose plugs. These were examined for signs of infection over a period of 21 days.

Results

Distribution of S. oenotherae pycnidia on seeds and capsules

Irregularly-shaped, fawn-coloured lesions of around 10 mm diameter containing developing pycnidia were commonly found on the rosettes of the inoculated plants. On flower spikes, disease symptoms were observed on many bracts and these subsequently spread to the capsules (Fig. 1). Lesions were also seen to start at the tip of the capsule. The spread of the disease was slow on immature green capsules but became more rapid as the capsules senesced. Lesions containing pycnidia also spread rapidly on the senescing stems with most of the stems being covered in pycnidia by November. Over 90% of pycnidia occurred on the outsides of the capsules with a significantly higher incidence ($T=2.46$, $p<0.05$) on the lower half of the capsule (Table 1). Only 0.4% were present on the seeds.

Internal infection by S. oenotherae in hand-harvested seeds

Similar data for the distribution of pycnidia were obtained in 1996 (Table 2). In the blotter test pycnidia only formed towards the end of the 21-day period as the seedlings were beginning to senesce. On PDA *S. oenotherae* recovered from seeds produced a dark colony with white edges which were 5–15 mm in diameter after 10–14 days. Pycnidia were malformed on PDA, based on the absence of an ostiole, and rarely contained distinct septate conidia. However, aseptically transfer of inoculum from these cultures onto healthy evening primrose seedlings or larger plants readily led to the development of typical disease symptoms that contained normal pycnidia within eight days. When plated on the medium developed for *S. nodorum*, a sparser, red-brown coloured colony grew from infected seeds. Again ostioles failed to form but septate conidia were found. Inoculation of healthy evening primrose seedlings again produced typical lesions of *S. oenotherae*. On dilute V8 agar, colonies similar in colour to those on PDA formed, with pycnidia again being malformed (Fig. 2). When *S. oenotherae* grew directly on seedlings germinating on the medium normal pycnidia were formed.

Infection by S. oenotherae in combine-harvested seed stocks

In the commercial seed sample no pycnidia were found on seeds or debris and no colonies of *S. oenotherae* were found on PDA (data not shown). Nineteen pieces of debris infected with pycnidia were found in the sample harvested from the fungicide trial control plots but no seeds bearing pycnidia were observed. After 21 days in the blotter test no seedlings infected with *S. oenotherae* were found and no colonies were formed on PDA.

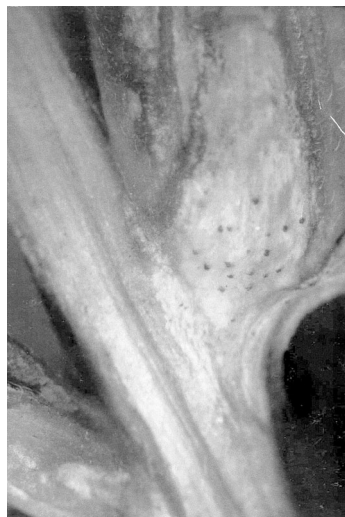


Fig. 1. Evening primrose (*Oenothera* spp.) bract with lesion and pycnidia associated with *S. oenotherae* infection that has spread to base of capsule

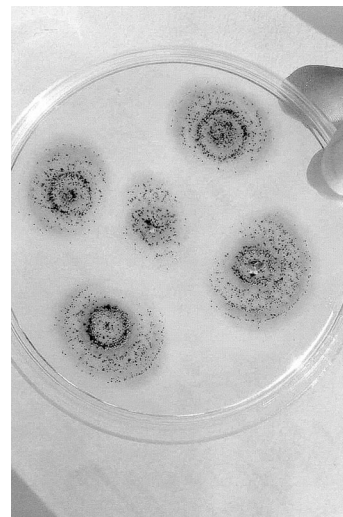


Fig. 2. Colonies of *S. oenotherae* developing from five evening primrose (*Oenothera* spp.) seeds after 14 days on dilute V8 agar. One seed has germinated and cotyledons are not yet showing symptoms of *S. oenotherae*

Table 1
Distribution of *S. oenotherae* pycnidia in 121 capsules of evening primrose (*Oenothera* spp.) harvested in 1995

Location	ESL	ESU	IS	Placenta	Seeds
Mean no. pycnidia per capsule	54.3	37.0	0.6	2.7	0.4
Standard error	5.3	4.6	0.2	1.0	0.2
% of total	57.2	39.0	0.6	2.8	0.4

ESL: External surface of capsule (lower half); ESU: External surface of capsule (upper half); IS: Internal surface of capsule

Table 2
Distribution of *S. oenotherae* in ten capsules of evening primrose (*Oenothera* spp.) harvested in 1996 and levels of internal seed infection

	No. of pycnidia on external capsule surface	No. of pycnidia on internal capsule surface	No. of pycnidia on placenta	No. of pycnidia on seeds	No. of infected seedlings/40 (Blotter Test)	No. of infected seeds/40 (PDA Test)
Mean	160.0	1.5	4.3	0	14.0	20.4
% of total	96.5	0.9	2.6	0		

Control of seed-borne S. oenotherae using hot water treatment

Using the blotter test *S. oenotherae* was detected in two (1997-harvested) seed stocks (Table 3) but not in the other eight (data not shown).

Seed stock 7 was used for the hot water treatment studies. The blotter test was repeated using 400 untreated seeds and similar data were obtained (Fig. 3). 87% of seeds had germinated after 14 days and three infected seedlings were detected after 21 days. 400 pycnidia-free seeds were then hot water treated at each of six temperatures and no pathogen infection was detected in any seed sample following hot water treatment.

In all treatments, few seeds germinated between 7 and 14 days and only the 14-day data are shown in Fig. 3. Seed germination declined steeply after immersion in water at temperatures greater than 50°C. A Kruskal-Wallis test confirmed that highly significant ($H=100$, $p<0.001$) differences in germination occurred across treatments, but a non-parametric Tukey-type multiple comparison test indicated that temperatures of 45 and 50°C caused no significant ($q=34.9$, $p<0.05$) reduction in germination.

Overwintering of S. oenotherae on aerial plant debris

Pycnidia containing abundant conidia typical of *S. oenotherae* could readily be found on overwintered dead stems and capsules. These conidia readily produced infection after 10–14 days when transferred to evening primrose seedlings growing on cylindrical cellulose plugs. Careful examination of this material over a number of weeks did not reveal a potential ascospore stage of this fungus.

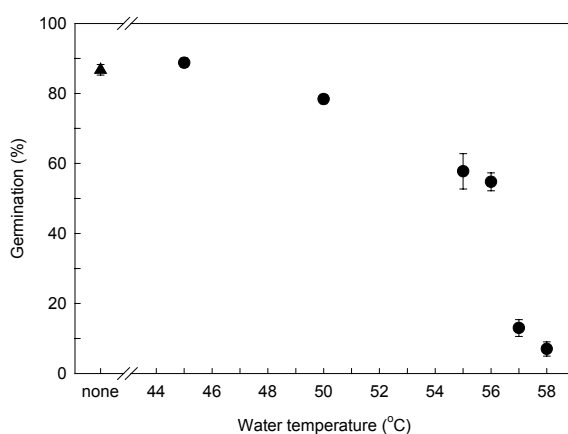


Fig. 3. Percentage seeds of evening primrose that had germinated after 14 days following soaking in hot water for 25 minutes at different temperatures (circles) and in the untreated control (triangle). Error bars represent ± 1 SE

Discussion

In the transplanted plants infection of capsules by *S. oenotherae* often followed infection of the subtending bracts and this may explain the higher number of pycnidia found on the lower half of the capsules. The number of pycnidia on the capsule surface increased markedly as the capsule ripened but few pycnidia were found on the internal capsule wall or on the placenta. Seeds bearing pycnidia accounted for only 0.4% of the pycnidia found on the 121 infected capsules examined (Table 1).

Internal infection by *S. oenotherae* was demonstrated not just in seeds exposed to an artificially high level of inoculum, but also, at a much lower level, in some commercial seed stocks (Tables 2 and 3). Although the incidence of infection in these stocks did not exceed 1%, at a sowing rate in commercial crops of approximately 800 seeds m^{-2} this represents an infection rate of up to 8 infected seeds m^{-2} . Stobart et al. (1999) have shown that even a low incidence of pycnidial infection (one seed per 2.5 m^2) can cause a significant increase in initial inoculum level. Thus, internal seed-borne infection cannot be ruled out as a potential primary source of infection in an otherwise healthy crop.

Hot water treatment at 45°C appears to be an effective method of controlling internal seed-borne infection by *S. oenotherae* in evening primrose. However, temperatures above 50°C caused a reduction in germination capacity (Fig. 1). A reduction in germination greater than 5–10% is considered unacceptable (Gratwick and Southey, 1986).

Table 3

Germination ability and levels of internal seed infection by *S. oenotherae* in seed stocks of evening primrose (*Oenothera* spp.) harvested in 1997

Seed stock No.	% germinated seed (7 days)	% germinated seed (14 days)	No. of infected seeds/400 (Blotter Test)
5	67	72	2
7	85	88	4

Strategies to reduce the incidence of *S. oenotherae* in commercial sowing stocks of evening primrose include the production of seed stocks in areas of low disease incidence (Simpson et al., 1995). In one such stock *S. oenotherae* could not be detected on either seeds or plant debris, nor was internal infection observed after plating the seeds onto agar.

Slow rosette growth during the winter months (Fieldsend and Morison, 2000a) means that crops are still susceptible to a low inoculum level if wet weather conditions prevail. The ease with which it has been shown that *S. oenotherae* can overwinter in aerial crop debris confirms that the agronomic practice of a rotation between evening primrose crops is a valuable precaution.

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